

NAME: _____

MATRIC NUMBER: _____

SCHOOL OF MATERIALS SCIENCE & ENGINEERING

CONTINUOUS ASSESSMENT 2 SEMESTER 1 2021-2022

MS1016 – THERMODYNAMICS OF MATERIAL

25 October 2021

Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains **THREE (3)** short questions and **TEN (10)** MCQ questions.
2. Answer **ALL** questions.
3. This is an open book test.
4. All questions have **ONLY ONE** correct answer for the MCQ questions.

PART I – SHORT QUESTIONS

- (1) Explain the difference between the ‘molar Gibbs free energy and the ‘partial molar Gibbs free energy’.

(20 marks)

Molar Gibbs free energy is the Gibbs free energy per unit volume of the material. It is calculated by dividing the total Gibbs free energy of a system by the number of mol. Hence the unit for the molar Gibbs free energy is Joules/mol.

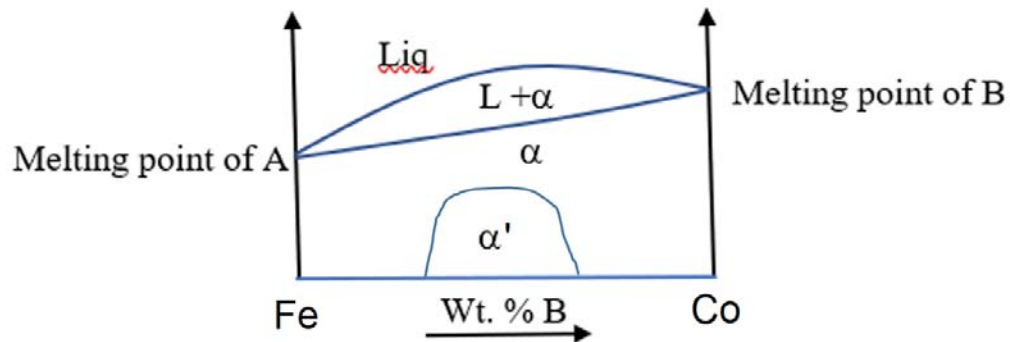
Partial molar Gibbs free energy is the partial derivative of the Gibbs free energy of a component in a mixture by the number of mole, keeping temperature and Pressure constant, and the other components in the mixture constant. It is thus defined as:

$$\left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_1,n_2,\dots,n_{j\neq i}}$$

The partial molar Gibbs free energy is also called the chemical potential of a component in a mixture. Partial molar Gibbs free energy has the same units of Joules/mole and represents the Gibbs free energy of the component in the mixture, taking into consideration the interaction with the other components.

2. FeCo is an important magnetic material used in electrical machine. Fe and Co is known to be totally miscible to each other and at lower temperature, forms an ordered structure. Sketch a likely representative of an equilibrium phase diagram to represent the Fe-Co system.

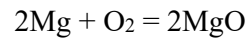
(20 marks)



In a binary mixture where the bond energy between the unlike bonds is stronger than the bonds between the same atoms/molecules, a homogeneous disordered solid phase, α is normally formed because the atoms are likely to be totally miscible to each other. The melting point of the mixture (alloy) tends to be higher than the respective pure components because of the more stable bonds.

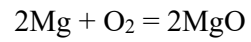
At low temperature, an ordered phase α' is formed as indicated in the phase diagram.

3. The Gibbs free energy for the formation of MgO at 1000 K is $-520,000$ J/mol. Calculate the equilibrium O_2 pressure for the oxidation of Mg at 1000 K. The oxidation reaction of Al is



(20 marks)

Given, the standard free energy for the formation of Mg is $-520,000$ J/mol



The equilibrium constant is

$$J = \frac{a_{MgO}^2}{a_{Mg}^2 a_{O_2}}$$

When the O_2 is at equilibrium with Mg, $J = \frac{1}{a_{O_2}}$, $\therefore \Delta G^0 = RT \ln P_{O_2}$

$$\therefore \Delta G^0(1000 \text{ K}) = -520,000 = 8.314 \times 1000 \ln P_{O_2}$$

$$\therefore \ln P_{O_2} = -62.5 \text{ atm}$$

$$\underline{P_{O_2} = 6.9 \times 10^{-28} \text{ atm}}$$

PART II MCQ QUESTIONS

- (4) Answer all the **TEN MCQ** questions below. There is only **ONE** answer for each question.

(40 marks)

- (i) What is the condition for equilibrium of two phases in a single component system?

- A the chemical potential of the phases must be equal
- B the molar Gibbs free energy of the phases must be equal
- C the total Gibbs free energy of the two systems must be constant
- D the mass of the two phases must be equal
- E the entropy of the two phases must be equal

ANS: B

- (ii) The Clausius-Clapeyron equation is given as:

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{\underline{S}^{\beta} - \underline{S}^{\alpha}}{\underline{V}^{\beta} - \underline{V}^{\alpha}} = \frac{\underline{\Delta S}}{\underline{\Delta V}}$$

The melting point of a certain solid was found to increase when the external pressure is increased. Which of the following deduction is probably correct?

- A dP/dT is positive
- B the material will melt easier under pressure
- C the volume of the material contracts upon melting
- D dP/dT is negative
- E None of the above

ANS: A

- (iii) Grain growth during heat treatment is the likely result of:

- A high entropy of the grains
- B high equilibrium vapour pressure of the grains
- C high atmospheric pressure of the grains
- D high vapour pressure of the grains
- E high surface energy of the grains

ANS: D

(iv) Which of the following elements cannot be oxidized in steel (Fe) making process (refer to Ellingham diagram)?

A Si

B Cr

C Ti

D Mn

E Cu

ANS: E

(v) The chemical potential of a component in a mixture is also called:

A partial molar Gibbs free energy

B partial molar volume

C partial molar enthalpy

D partial molar entropy

E partial molar heat capacity

ANS: A

(vi) In the extraction of Si from SiO_2 , from metal oxide, low equilibrium pressure of O_2 can be overcome by the following methods, EXCEPT:

A introducing a vacuum environment

B introducing hydrogen

C increasing oxygen pressure

D introducing CO

E introducing Al

ANS: C

(vii) What is an intermetallic?

- A A compound of a metal and oxygen
- B a crystal consisting of 2 metallic atoms randomly arranged
- C a mixture consisting of 2 metallic atoms
- D metal dissolved in another metals
- E a crystal consisting of 2 metallic atoms whose crystallographic positions are fixed

ANS: E

(viii) Refer to the Ellingham diagram. Ti and Si are possible candidates to be used to remove oxygen from molten Fe at 1500°C. Which element is more efficient __**(i)**__? Explain Why_____**(ii)**_____.

- A (i) Ti (ii) higher equilibrium O₂ pressure
- B (i) Ti (ii) lower equilibrium O₂ pressure
- C (i) Si (ii) higher equilibrium O₂ pressure
- D (i) Si (ii) lower equilibrium O₂ pressure
- E (i) both Si and Ti are equally effective (ii) similar equilibrium O₂ pressure

ANS: B

(ix) Which of the following equilibrium is not possible?

- A 4 phases in equilibrium in a binary alloy
- B 3 phases in equilibrium in a binary alloy
- C 2 phases in equilibrium in a binary alloy
- D 2 phases in equilibrium in a single component
- E 3 phases in equilibrium in a single component

ANS: A

- (x) Figure (X) below is a binary phase diagram. The solubility limit of B in A labelled in the phase diagram is represented by the letter:

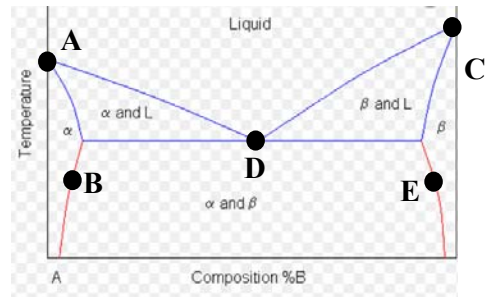


Figure X

ANS: B

